Investify

A personalized stock investment advisor that reasons over point-in-time data and backs every recommendation with a SHAP-explained rationale.

Supervisor

Mr. Irfan Ullah

Members

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Problems and Users

End users: Investors using AI-powered advisory platforms

Current Workflow: Users get a Buy/Hold/Sell call with little insight into why and no guarantee the model wasn’t trained on future information

Why it matters: Temporal data leakage makes models look accurate in back-tests but may fail in live trading

Existing Solutions

Product Comparison Table

Feature	Robo-Advisors eg Wealthfront	AI Stock Scoring Apps	Sentiment-only tools eg StockTwits	Professional Terminals eg Bloomberg	Published FinBERT + SHAP work	Investify
Personalized risk profiling	Basic questionnaire	✗	✗	✗	✗	✓ Capacity + Grable–Lytton scale
Formal + informal sentiment fusion	✗	Formal news only	Informal only	✓	News only Single register	✓ FinBERT + CryptoBERT
Explainable predictions	✗	Generic score only	✗	Not inspectable	Tree SHAP + unverified	✓ Tree SHAP + verified
Leakage-safe evaluation	✗	✗	✗	Not disclosed	Calendar date align only	✓ Walk forward + point in time
Natural-language explanation	✗	✗	✗	✗	✗	✓ LLM explanation layer

R&D and Technical Aspect

Source	Key Finding	What we learned
Stock Price Prediction Using FinBERT-Enhanced Sentiment with SHAP Explainability - Ruan & Jiang (Mathematics, 2025)	Combines FinBERT sentiment with technical indicators in XGBoost; SHAP shows consistent, interpretable feature contributions	Validates our FinBERT + XGBoost + SHAP pipeline choice
Profit Mirage: Revisiting Information Leakage in LLM-based Financial Agents - Li et al. (arXiv 2510.07920, 2025)	Back tested LLM trading returns collapse once models face genuinely post-training data	Directly justifies our walk-forward and negative-control evaluation design
When Alpha Disappears – decision-time leakage benchmark for financial back-tests (2026)	Leakage isn't binary. even small protocol conventions (e.g., using same-day close) inflate backtest results	Motivates our strict point-in-time cutoff enforcement, not just walk-forward splitting alone

Complex Computing Problem

1) **Leakage-safe pipeline design:** designing a leakage-safe ML pipeline, multi-model orchestration not just calling an API

2) **Multi-model orchestration:** Reconciling sentiment models, XGBoost, SHAP, and an LLM as one coherent system, not isolated scripts.

3) **Sentiment reconciliation:** fusing two independent models (formal news vs. informal social) into a single, coherent feature space

4) **Calibrated explainability:** translating SHAP attributions into natural language via LLM *without* the LLM inventing its own reasoning

Proposed Solution

Proposed Workflow

Risk Capacity + Tolerance -> Multi-source sentiment (FinBERT + CryptoBERT)

-> XGBoost prediction -> SHAP attribution -> LLM explanation layer

Modules

1. Investor Profiling
2. Sentimental Analysis
3. Predictive Model
4. Explanation Verifier
5. Explainability
6. Conversational Interface

Our claim

What we do not claim

No novelty in sentiment scoring, gradient boosting, SHAP-to-text, or QLoRA

These all are engineering inputs. And no profit claim of any kind.

What we claim

1. A leakage-audited point-in-time evaluation harness.
2. A deterministic verifier for generated explanations.
3. Cross-register divergence features with a pre-registered ablation.

Contribution

A leakage-audited, point-in-time harness combined with a deterministic explanation verifier, delivering a risk-personalized advisory pipeline for selected S&P 500 stocks

Selected Stocks

1. AAPL (Apple)
2. TSLA (Tesla)
3. NVDA (NVIDIA)
4. MSFT (Microsoft)
5. META (Meta/Facebook)
6. AMD (Advanced Micro Devices)
7. PLTR (Palantir)

Proof of Concept

Evaluation Plan and Risks

Success Metrics

1. Walk-forward backtest accuracy / macro-F1
2. leakage-control comparison (walk-forward vs. shuffled-fold)
3. calibration quality

Risks

1. Sentiment data sparsity for smaller tickers
2. LLM explanation drift from SHAP values
3. API rate limits

Work Division & Iterations

Mid Defense

Prove core technical building **blocks work in isolation**

Complete Risk Profiling Layer

Historical Data Pipeline

FinBERT and CryptoBERT scoring on sample data set

Final Defense

Functional web prototype: risk quiz → company selection → recommendation → SHAP-based explanation

XG model trained

SHAP integrated

LLM explanation layer

Nightly batch sentiment scoring pipeline operational

Mid Evaluation

All modules working together as **one coherent, stable system**

Full integration across all 7 tickers (not just baseline ticker)

Chat/follow-up interface with 10-message rolling context window implemented and tested

Leakage evaluation completed: walk-forward vs. shuffled-fold comparison, both negative controls (1-day horizon, shuffled-fold) run

Performance/latency check

Final Defense

Production-ready deployment and complete documentation

Complete project report: methodology, architecture, evaluation, limitations, future work

Work Division

Ahmed Javaid

point-in-time harness

leakage tests

dual-timestamp schema

parity test

deployment

Rafay Ul Haq

risk profiling

sentiment pipeline

degraded operation

acceptance testing

frontend

Soban Ahmed

decision engine

Calibration

SHAP

Verifier

QLoRA distillation

DEI, Ethics and Legal Issues

1) **DEI:** Risk assessment (capacity + tolerance) designed to be accessible regardless of financial literacy background

2) **Ethical:** Not licensed financial advice, not for live trading; explanations must not overstate model certainty; sentiment inputs disclosed as manipulable

3) **Legal:** Compliance with data source terms (Reddit/StockTwits API usage); no non-public/insider information; personal financial data stays local to deployment

Thank you.

Any Questions?